

Labwork 2

HPC – Get to know your GPU

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1 Knowledge check

I learned how to use CUDA for GPU computing in Python. I used `is_available()` to check whether CUDA is available. I used `list_devices()` to find available GPUs and `select_device()` to choose one. The Device provides the GPU's name, ID, compute capability... The Context provides information about the GPU's memory...

2 GPU information

Table 1: GPU information reported by `numba.cuda` on Google Colab.

Property	Value
Device name	Tesla T4
Compute capability	7.5
Multiprocessor count	40
Cores per SM	64
Total CUDA cores	2560
Total memory	14.56 GB